

Datathon 2026 - Urban Flow Analytics Data Challenge

Field Reference Guide

This reference guide describes the fields included in the datasets provided for this Datathon. Use it to understand what each column represents before building your models.

Urban Flow Analytics Taxi Dataset

Column Name	Description
provider_code	Numeric identifier for the technology company that supplied the trip record (e.g., specific mobile dispatch/metering providers). 1 = Creative Mobile Technologies, LLC 2 = Curb Mobility, LLC 6 = Myle Technologies Inc 7 = Helix
pickup_timestamp	Date and time at which the taxi's meter was switched on to begin the trip.
dropoff_timestamp	Date and time at which the taxi's meter was switched off at the end of the trip.
rider_count	Count of riders traveling in the vehicle during the trip.
distance_miles	Total distance covered during the trip, as measured by the vehicle's meter, in miles.
rate_class_id	Code representing the rate applied to the trip by the time it ended. 1 = Standard rate 2 = JFK 3 = Newark 4 = Nassau or Westchester 5 = Negotiated fare 6 = Group ride 99 = Null/unknown
offline_record_flag	Indicator showing whether the trip data was temporarily saved on the vehicle's onboard system before being transmitted, due to a lost connection with the central server.
origin_loc_id	Identifier for the geographic location where the passenger was picked up (meter activated).

Column Name	Description
dest_loc_id	Identifier for the geographic location where the passenger was dropped off (meter deactivated).
fare_settlement_method	Code describing how the fare was settled by the customer (e.g., card, cash, disputed charge, voided, or unknown method). 0 = Flex Fare trip 1 = Credit card 2 = Cash 3 = No charge 4 = Dispute 5 = Unknown 6 = Voided trip
base_fare	The core charge computed from time and distance traveled, per the meter.
surcharge_misc	Additional fees and surcharges applied outside the base fare.
transit_tax	A fixed tax applied automatically depending on the metered rate used.
driver_tip_payment	Recorded gratuity, but only reflects tips added via card; cash gratuities are excluded.
toll_total	Sum of all toll charges incurred during the trip.
service_improvement_fee	A flat surcharge applied at the start of every trip, introduced in 2015 to fund service improvements.
charge_total	Full amount billed to the passenger, excluding any cash-based tips.
zone_congestion_fee	Fee collected per trip to support New York State's congestion pricing initiative.
airport_pickup_fee	Flat fee applied only when the pickup occurs at one of the city's two major airports.
congestion_relief_fee	Charge levied per trip within a designated congestion relief zone, effective from early January 2025 onward.

Urban Flow Analytics Zone Dataset

Column Name	Description
loc_id	Unique numerical identifier assigned to each distinct taxi zone. Maps directly to origin_loc_id (pickup) and dest_loc_id (drop-off) in the main dataset.
borough_name	The administrative district or major administrative division of the city in which the taxi zone is located.
zone_name	The specific neighborhood or landmark name corresponding to the loc_id.
service_zone	Service classification area used to determine specific taxi operating regulations, pricing structures, and pickup permissions.